

An Explainable AI Framework for Academic Article Classification to Support Research and Learning

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Introduction

- Scientific publications are growing rapidly.
- Automatic article classification helps researchers organize and discover relevant literature.
- SciBERT classifies scientific text accurately but operates as a black box.
- The lack of explanation reduces researchers' trust in its predictions.
- Explainable AI (XAI) methods, such as SHAP and Integrated Gradients, improve transparency while maintaining accuracy.

Problem Statement

- Transformer-based classifiers achieve high accuracy but provide no explanation for their predictions.
- Most XAI studies rely on a single explanation method with limited human-readable output.
- Explanation faithfulness is rarely evaluated rigorously.
- Few studies assess both explanation fidelity and user trust.
- Our pilot study (n = 100) found that **83%** struggled to identify research categories, while **86%** preferred an explainable classification system.

83%

found category identification difficult

86%

wanted an explainable classification system

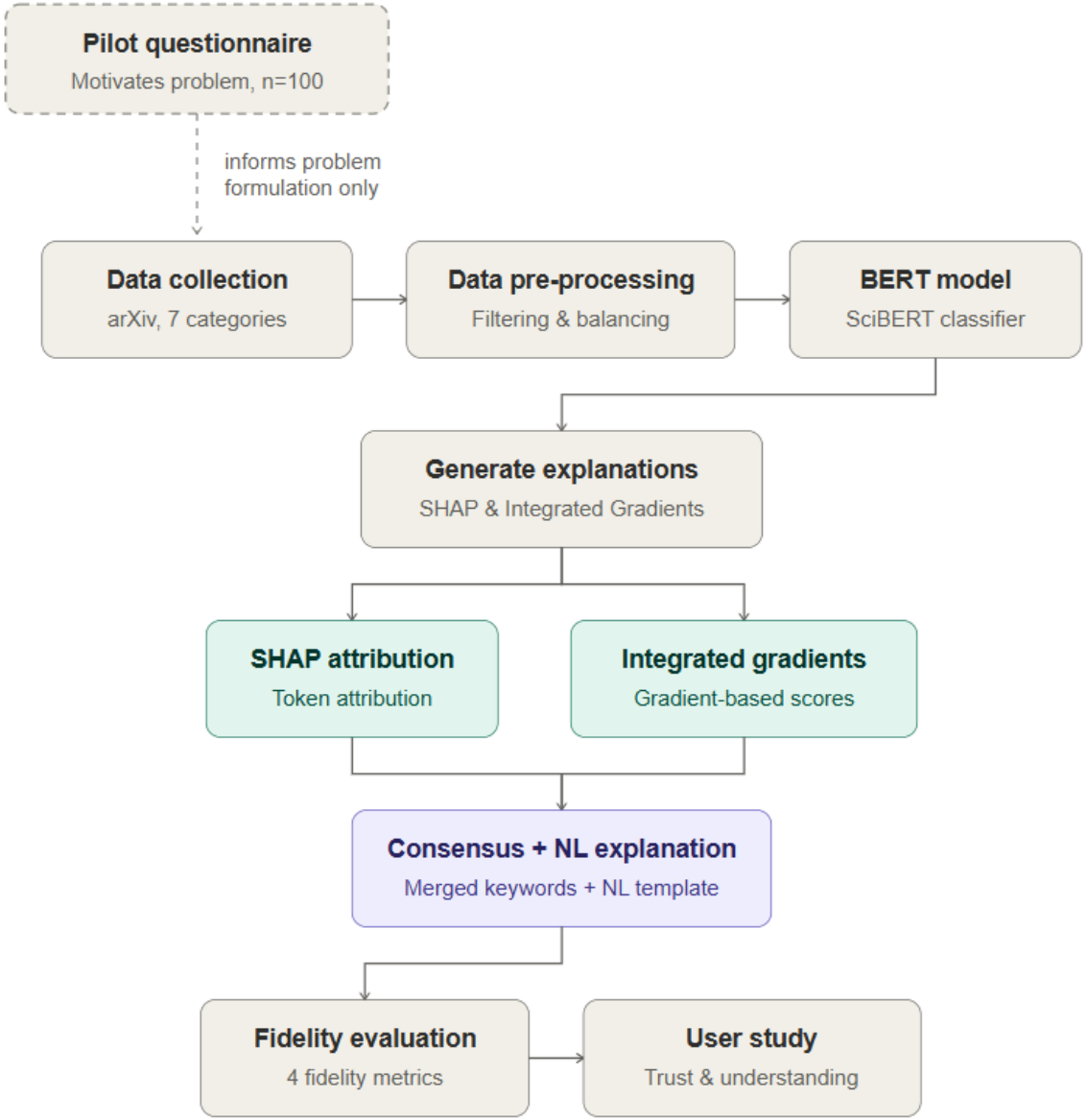
Literature Review

Study	Year	Focus	Method	Key Limitation
Mersha et al. [1]	2025	Evaluate XAI techniques on encoder-based LMs	Comparative evaluation of SHAP, IG, and attention	Compares methods individually — never tests whether combining them improves faithfulness
Atanasova et al. [2]	2020	Diagnose faithfulness properties of explainability techniques	Diagnostic comparison framework (LIME, saliency maps, gradient-based methods)	No single method dominates across all faithfulness diagnostics; fusion not explored
DeYoung et al. [3]	2020	Benchmark rationalized NLP models	ERASER (sufficiency & comprehensiveness metrics)	Faithfulness metrics developed and validated on general-domain NLP, not scientific/academic text

Objectives

- To investigate challenges in CS article classification and perceptions of AI-based classification.
- To develop and evaluate a fine-tuned SciBERT model for 7-category article classification.
- To compare SHAP, Integrated Gradients, and a SHAP–IG consensus approach for explanation faithfulness.
- To evaluate the impact of explanations on user trust and understanding.

Methodology



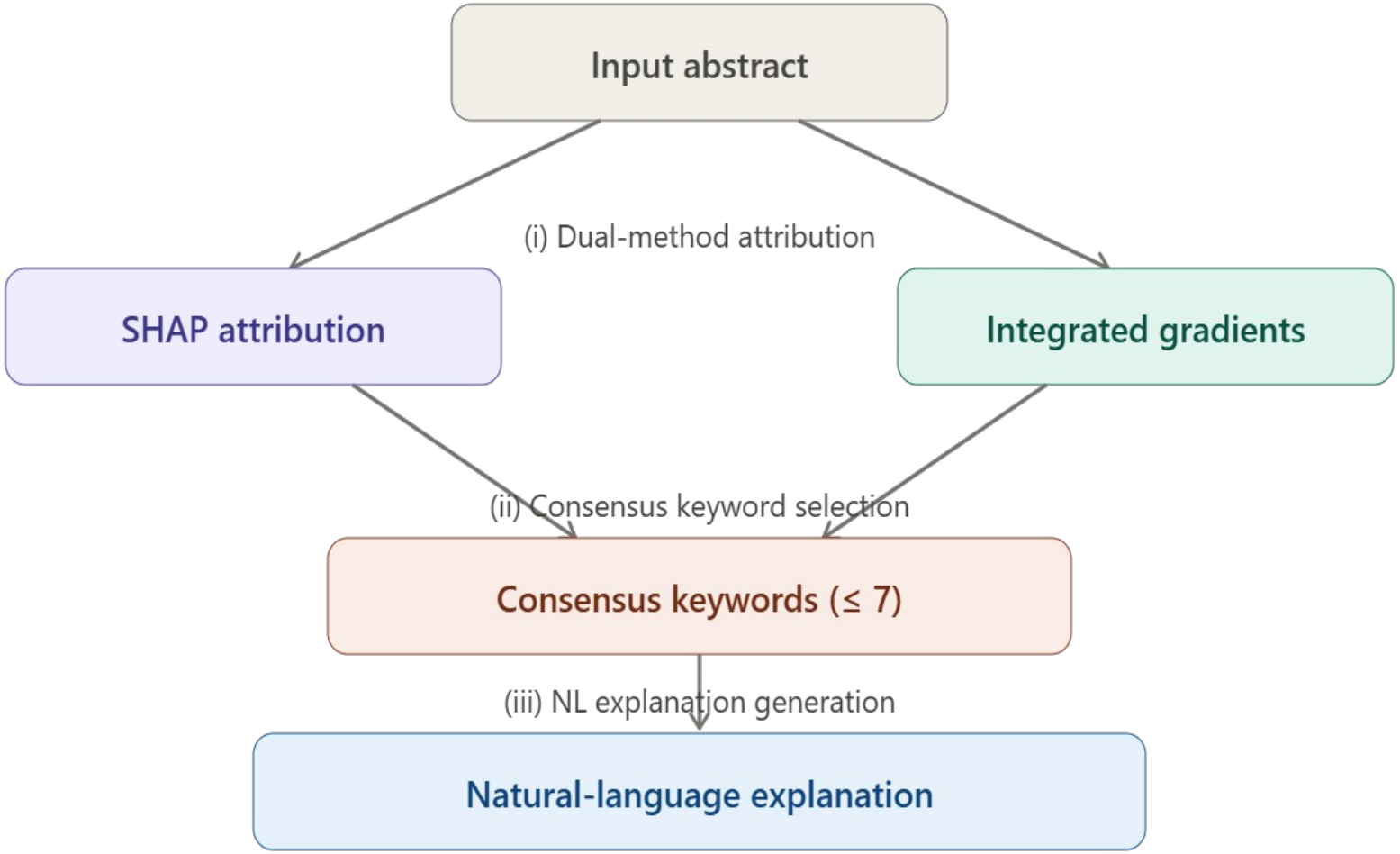
Dataset & Classification Model

- **Dataset**
  - 35,000 arXiv abstracts, 7 CS categories (LG, AI, CV, CL, RO, CR, CY), 5,000 per category
  - Single-label papers only; title + abstract as input
  - Stratified 70:15:15 train / validation / test split

- **Model**

  SciBERT (allenai/scibert_scivocab_uncased) fine-tuned with linear classification head
  - WordPiece tokenizer, max sequence length 512
  - 5 epochs, learning rate 2e-5, batch size 16, fp16, label smoothing

Explanation Framework



Fidelity Evaluation

- 35 fully held-out papers, never seen in training/validation/testing
- Four metrics: sufficiency, comprehensiveness, flip rate, stability
- Computed identically for SHAP, IG, and consensus for direct comparison

Metric	How it's measured
Sufficiency	Keep only explanation keywords → same prediction?
Comprehensiveness	Remove keywords → confidence drop?
Stability	Reword input → keyword set stays same?

Classification Results

Category	Precision	Recall	F1-score
ML	0.89	0.77	0.83
AI	0.93	0.95	0.94
CR	0.93	0.98	0.95
CY	0.91	0.90	0.91
CV	0.90	0.92	0.91
NLP	0.82	0.85	0.83
RO	0.94	0.95	0.95

Overall: 90% accuracy, 0.90 weighted F1

Main confusion pairs: ML ↔ NLP and CY ↔ NLP — categories with genuinely overlapping vocabulary

Explanation Fidelity Results

Method	Sufficiency (%)	Comprehensiveness	Stability (%)
SHAP	80.6	0.241	98.9
Integrated Gradients	80.9	0.267	99.1
SHAP + IG Consensus	**82.9**	**0.31**	**100**

The consensus method outperformed both individual methods on all fidelity metrics

User Study Results

N = 300 participants, pre/post design

Dimension	Pre	Post	Change
Trust (TiA)	3.15	3.96	**+0.81**
Usefulness (TAM)	3.41	4.14	**+0.73**
Interpretability	—	4.08/5	post only

Discussion

- Researchers struggle with article classification and prefer explainable predictions.
- Most errors occurred between similar categories (e.g., Computer Vision and Machine Learning).
- SHAP–Integrated Gradients produced more faithful and stable explanations.
- Improved trust and usefulness support literature review and research discovery.
- A trade-off exists between speed and explainability: 17 ms for classification versus 6.1 s for explanations.
- Limitation:The framework is currently limited to Computer Science articles.

Conclusion & Future Work

Conclusion

- Developed an explainable SciBERT framework for academic article classification.
- Achieved 90% classification accuracy with faithful explanations.
- Improved trust, perceived usefulness, and interpretability.

Future Work

- Evaluate the framework with expert researchers.
- Extend to multi-label classification.
- Apply the framework to additional scientific domains.

Thank you for your attention.